

Shanu Kumar

Senior Platform Engineer | Cloud Infrastructure | Site Reliability Engineering

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Senior Platform Engineer with 5+ years of experience designing, scaling, and operating high-availability cloud-native infrastructure, Internal Developer Platforms (IDP), and site reliability solutions on AWS and Kubernetes. Specialist in Infrastructure as Code (OpenTofu, CDKTF), service mesh architecture (Istio Ambient, Gateway API), KEDA autoscaling CI fleets, and AWS Bedrock platform automation. Proven track record of empowering 20+ engineering teams, driving enterprise cloud cost efficiency, and accelerating developer delivery.

TECHNICAL SKILLS

- **Cloud Infrastructure & IaC:** AWS (EKS, EC2, Lambda, IAM, RDS, VPC, Transit Gateway, Route 53, S3), OpenTofu, Terraform, CDKTF, Crossplane
- **Containers & Service Mesh:** Kubernetes, Istio Ambient Mesh, Gateway API, NLB / cert-manager, Docker, Helm, KEDA, Karpenter, vCluster
- **Developer Platform & CI/CD:** Backstage, ArgoCD, Azure DevOps, GitHub Actions, GitLab CI, Actions Runner Controller (ARC)
- **Observability & Security:** OpenTelemetry, Prometheus, Grafana, Loki, Tempo, Datadog, Dependency-Track (SBOM), Okta IAM, Kong API Gateway
- **Programming Languages:** TypeScript, Python (boto3), Bash, SQL
- **AI Platforms & Automation:** AWS Bedrock, RAG Architecture, Model Context Protocol (MCP), Agentic Workflows

EXPERIENCE

Lead Engineer

Verisk Analytics

Apr 2024 – Present

Hyderabad, India

- **Developer Platform Architecture:** Architected an Internal Developer Platform (IDP) for 20+ engineering teams; built a Backstage-integrated Stack Management Platform (TypeScript, DynamoDB, CDKTF) enabling self-service infrastructure, ownership indexing, and Terraform state governance.
- **CI/CD Infrastructure & Cost Optimization:** Migrated CI/CD execution from Azure DevOps cloud runners to self-hosted EKS build fleets using KEDA, cutting infrastructure costs by 30% and achieving 80% faster runner startup times.
- **Service Mesh & Traffic Management:** Led zero-downtime migration of 100+ services from Kubernetes Ingress to Gateway API with Istio Ambient mode and NLB/cert-manager, eliminating per-pod proxy overhead and cutting CPU/memory usage by 25%.
- **Infrastructure as Code Governance:** Built modular OpenTofu & CDKTF blueprints for Amazon RDS and Kubernetes resources, reducing environment provisioning time by 90% and standardizing cloud deployments.
- **AI Automation, Security & Observability:** Engineered an AI runbook assistant (AWS Bedrock RAG, Claude 3.5 Haiku), integrated Dependency-Track for SBOM security, and implemented OpenTelemetry LGTM tracing across Kubernetes & AWS Lambda.

Technical Lead

HCL Technologies

Oct 2022 – Mar 2024

Pune, India

- **AWS Multi-Account Architecture:** Architected multi-account AWS topology using Transit Gateway, IAM role federation, and cost governance strategies, reducing cloud infrastructure expenditure by 40%.
- **CI/CD & IaC Standardization:** Designed multi-cloud CI/CD automation pipelines and reusable Helm/RDS Terraform modules, reducing release cycle times by 50% and configuration overhead by 85%.
- **Kubernetes Autoscaling & Resilience:** Implemented Karpenter and HPA autoscaling policies across multi-tenant EKS clusters, optimizing node utilization and ensuring high availability for core services.

- **GitOps & Microservices Infrastructure:** Deployed 50+ production microservices using ArgoCD GitOps, Helm, RBAC, HPA, and NetworkPolicies, establishing Blue-Green deployment strategies for zero-downtime releases.
- **Stateful Distributed Systems:** Designed and operated highly available Hazelcast CP clusters on Kubernetes using StatefulSets, ensuring strong consistency, resilience, and fault tolerance for stateful applications.

PROJECTS

- **Multi-Tenant Internal Developer Platform (PaaS):** Engineered a 2-layer EKS v1.35 PaaS separating OpenTofu core infrastructure from ArgoCD GitOps, Crossplane XRD compositions, vCluster tenant isolation, and Keycloak authentication.
- **Autonomous SRE Alert Remediation Agent:** Built a multi-agent incident remediation platform using AWS Bedrock (Claude 3.5 Haiku), DynamoDB, and 8 Model Context Protocol (MCP) servers with human approval.
- **Platform Observability Stack & SLO Engine:** Architected a Kubernetes telemetry stack covering 4 golden signals using Prometheus, Loki, Grafana, OpenTelemetry, and 38 custom AWS/K8s alert rules.
- **RAG Knowledge Assistant for Engineering Runbooks:** Implemented an async ingestion pipeline (S3, Lambda, EventBridge) with AWS Bedrock Knowledge Base, OpenSearch Serverless hybrid search (k-NN + BM25), and Kong API Gateway.

EDUCATION

Master of Technology (M.Tech), Computer Science & Engineering Indian Institute of Technology Madras (IIT Madras)	Jul 2019 – Jul 2021 Chennai, India
Bachelor of Technology (B.Tech), Computer Science & Engineering Loknayak Jai Prakash Institute of Technology	Sep 2014 – Jul 2017 Chapra, India

CERTIFICATIONS

- Certified Kubernetes Administrator (CKA, CNCF)
- Certified Kubernetes Application Developer (CKAD, CNCF)
- HashiCorp Certified: Terraform Associate
- AWS Certified AI Practitioner
- AWS Certified Cloud Practitioner

AWARDS

- AWS GameDay Cloud Architecture Competition – 2nd Runner-up (Jul 2023)
- ST Cadre Star Performer Award (2023)
- HCL Distinguished Contributor Award (2023)